and is just one click away. Patients can also share their data with researchers and help improve research on many critical illnesses. It also solves the counterfeit drug problem — helping both patients and drug companies.

Blockchain architecture for beginners

Now that we have some idea of how a blockchain functions, it is time to learn about its architecture.

The key components of the blockchain architecture are:

- Transactions
- Blocks
- Mining
- Consensus

Apart from the general components, there are different types of blockchain architecture like public, private, and consortium.

Blocks: A blockchain is composed of blocks. The blocks are stored in a linear fashion whereby the latest one is attached to the previous block. Each block contains data — the structure of the data stored within the block is determined by the type of blockchain and how it manages the data.

Let us look at the example of the Bitcoin blockchain. A block in Bitcoin contains the basic information about a transaction, including the receiver, the sender, and the amount of Bitcoin transferred. The first block of any blockchain is known as the genesis block. This is the only one that doesn't have any preceding block.

In a block, there is important information known as the hash. This is used to determine the authenticity of any block and whether it should be attached to the current chain. The hash is unique to every block and hence cannot be replicated by any malicious block. It is also a gateway to understanding what the block includes. This enables the block to protect its contents. So, if someone tries to change the information within the block, the hash value will also change, triggering a warning so that other blocks do not accept it.

The structure of each block can be divided into three parts — the data, hash, and the previous block hash. The whole chain is constructed in this manner.

Transactions: A transaction takes place within the network when one peer sends information to another. It is a key element of any blockchain and without it there would be no purpose in initiating a transaction. A transaction consists of information, including the sender, the receiver and the value. It is similar to a transaction done on modern credit card platforms. The only difference is that the transaction here is done without a centralised authority.

A simple example would be of one user sending Bitcoin to another. The transaction initiates an agreed-contract blockchain which changes its state. As the whole blockchain is a decentralised network, it needs to be updated by all the nodes. Each node contains an exact copy of the ledger, and thus, a state of blockchain is created. Any single transaction can initiate a state change.

As mentioned earlier, a block contains a bunch of transactions. There is a limit to how many transactions a block can contain. This depends on the block, the transaction size and also on the limit imposed on how many transactions can stay in a block. The verification of the transaction is done by independent nodes based on the consensus method used. Technically, each transaction can have one or more input(s) and output(s). In this way, the transactions are linked so that they can keep a proper note on the expenditure done in the blockchain.

Mining: Mining is an important part of a blockchain network. Bitcoin uses mining for the proof-of-work (PoW) state. Clearly, block creation requires real-world effort, which is mining. The mining is done by spending computational units to solve complex mathematical puzzles.

The miners are responsible for mining operations. Nodes with the necessary hardware participate in the mining process. They are required to solve complex mathematical solutions and hash the block. They are also incentivised for doing the hard work. Their payment, however, depends on the mining difficulty level and the work done towards validating a block.

Mining is only required in public blockchains.

Consensus: The last important part of blockchain architecture is consensus. This is the method by which a transaction is validated. Each blockchain can have a different consensus method attached to it. For example, Bitcoin uses proof-of-work (PoW), whereas Ethereum uses proof-of-stake (PoS). There are other types of consensus methods as well.

A consensus method is a set of rules that needs to be followed by everyone in the network. Also, to impose a consensus method, the nodes should participate. Without any node participation, the consensus method cannot be implemented. This also means that the more nodes that join in to participate in the consensus method, the stronger the network becomes. Bitcoin has a big network and offers great incentives to become a miner. In fact, it also has one of the biggest miner communities out there.

The miners can collectively also follow their own wishes or make their point. For example, if there is a change required in the blockchain, then miners can decide on how to achieve that change. Conversely, if a change that they do not approve of has been made, they can protest against it. Miners or the nodes taking part in the consensus method are also capable of hijacking the network if more than 51 per cent of them are controlled by any one entity. This is known as a 51 per cent attack, whereby more than half the nodes are controlled by one entity. In such cases, the nodes can fake